



SIMATS ENGINEERING

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Course Code: CSA1522

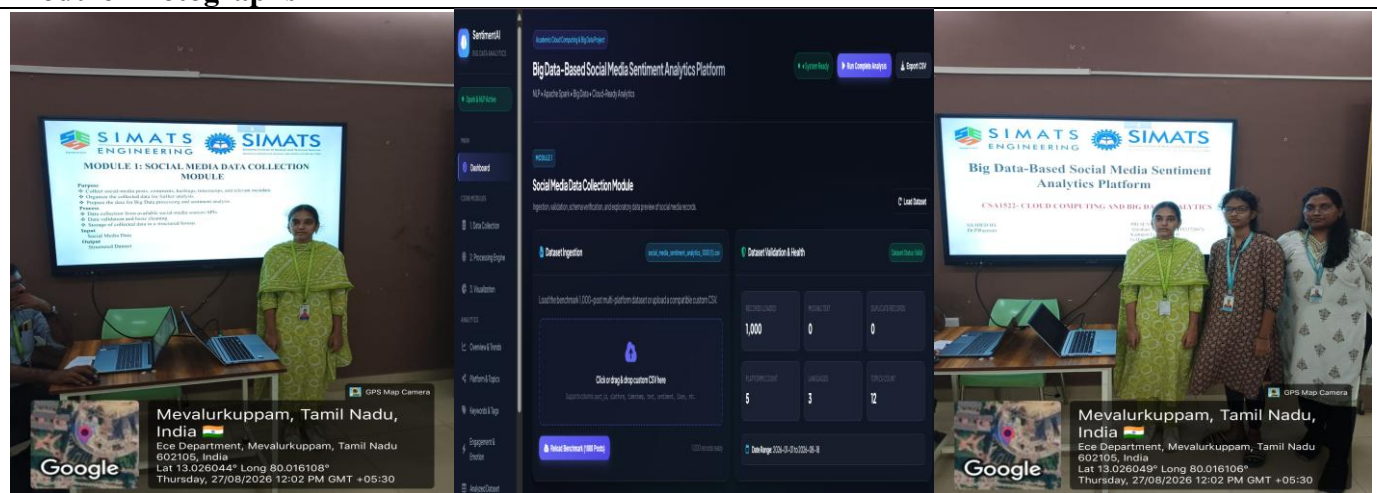
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Course Name: Cloud Computing and Big Data Analytics

Course Faculty: Dr. P Rajaram

Project Title: Big Data-Based Social Media Sentiment Analytics Platform

Module Photographs:



Project Description:

The Big Data-Based Social Media Sentiment Analytics Platform is a data analytics system developed to collect, process, and analyze the massive amount of information generated from social media platforms, including posts, comments, reviews, and user feedback. Since social media data is continuously generated in large volumes and is mostly unstructured, manually analyzing it is difficult and time-consuming. The proposed platform combines Big Data processing, Natural Language Processing (NLP), and Sentiment Analysis to automatically transform this raw social media content into meaningful insights. The core Sentiment Analysis & Big Data Processing Engine receives the collected data and performs pre-processing operations such as removing duplicate records, unnecessary characters, URLs, and irrelevant content, followed by text normalization and analysis. Apache Spark is used to process large-scale datasets efficiently through distributed computing, providing faster and scalable data processing. NLP techniques are applied to understand the textual content, after which the system classifies user opinions into Positive, Negative, or Neutral sentiments. The engine also aggregates the analyzed results to identify overall sentiment, sentiment percentages, topic-wise opinions, and changing trends in public reactions. These results can be further represented through charts, graphs, and dashboards to provide an easy-to-understand view of social media trends. The platform can be useful for brand monitoring, customer feedback analysis, market research, product evaluation, public opinion analysis, and data-driven decision-making, providing organizations with a scalable solution for converting large volumes of social media data into actionable insights.

Student Signature

Guide Signature